

# MARKO CIRIC

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## EDUCATION

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**University of Toronto | Bachelor of Applied Science, Electrical and Computer Engineering** Toronto, ON

Dean's Honours List (2020F, 2021S) 2020-2025

## TECHNICAL SKILLS

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**Software:** C, C++, Python, Perl, ARM Assembly, MATLAB, Unix, Git, Perforce, Vim

**Hardware:** SystemVerilog, Intel Quartus Prime, ModelSim, Synopsys VCS, Synopsys Verdi, Lint, FPGA, ASIC, Altium Designer, NI Multisim, LTspice, Synopsys Taurus TSUPREM-4 TCAD, Cadence Virtuoso Layout Suite

## EXPERIENCE

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**Graphics IP RTL Design Intern • AMD • Co-op** May 2023 – Present

- Working on the **Geometry team** as an RTL designer for front-end Graphics pipeline to implement tessellations for multiple next generation high performance Radeon Graphics Cards.
- Providing day-to-day support for the design team by resolving triage failures using waveform debugging.
- Working on implementing new features, standardizing interfaces using common modules, and improving legacy code.

**UTFR Driverless Vehicle Development Hardware Team Member** August 2022 - April 2023

- Worked as a member of the **University of Toronto Formula Racing Team** on developing the electrical activation of the emergency brakes (EBS), designing a controller to receive/transmit data over the CAN bus and distribute power to the driverless vehicle's electrical devices.

## TECHNICAL PROJECTS

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**Wireless Communication between STM32** April 2023 - April 2023

Completed as part of design course in group of 2: ECE342: Computer Hardware

- Using two STM32F4s with antennas (for mobility), with one connected to a transmitter and the other to a receiver that receives data through wireless transmission with DMA and custom encoding/decoding of the transmission.
- Data was transmitted accurately within distances of approximately 15 meters; but able to reach as far as 20 meters.

**Power Amplifier and Filter System in Software-Defined Radio (SDR)** January 2022 – April 2022

Completed as part of design course in group of 3: ECE295: Hardware Design and Communication

- Designed a class-B discrete transistor power amplifier with a (LC) low-pass filter for passing the signal in the 8-16 MHz (RF) range to drive the SDR antenna with requirements of 1 W output power and a THD of less than 10%.
- Tested functionality of the design through simulations on NI Multisim achieving 1.33 W output power and THD ~0.3%.
- Created electrical schematic in Altium Designer passing the ERC and completed PCB design process passing the DRC.

**Custom Processor Design** March 2022 – April 2022

Completed as part of design course in group of 2: ECE243: Computer Organization

- Designed a 16-bit, 8-register, "ARM-like" processor in Verilog capable of basic instructions, subroutine, stack functionality and with connections to external devices with memory mapped I/O (SWs, KEYS, LEDRs, etc.).
- Simulated and debugged the processor's Verilog using ModelSim to test functionality of the instructions.